Chapter 16
CFD Modelling of Mortar Extrusion and Path Planning Strategy at the Corner for 3D Concrete Printing

Khalid El Abbaoui, Issam Al Korachi, Md Tusher Mollah, and Jon Spangenberg

Abstract Computational Fluid Dynamics (CFD) modelling has become an important tool for accelerating different costly and time-consuming processes for 3D concrete printing. In this study, a CFD model is developed to simulate the mortar extrusion and deposition flow in order to investigate the path planning at the corner. The mortar is characterized by a viscoplastic Bingham fluid. The printhead moves according to a prescribed speed to print trajectory with a turn of 90° angle. The model solves the Navier Stokes equations and uses the volume of fluid (VOF) technique. This work investigates the effects of changing printhead direction in the geometrical conformity and the process precision. The direction change is carried out by the acceleration steps and jerk. A smoothing factor is provided to smooth the toolpath. Several simulations were performed by varying the smoothing factor and jerk. Overfill has occurred at the sharp corner when applying a jerk due to the constant extrusion rate. Dependance between the acceleration steps, the acceleration time limit and the jerk has been found.

Keywords 3D concrete printing · CFD modelling · Path-planning

16.1 Introduction

The automated construction process, 3D concrete printing (3DCP), has gained significant attention over recent years due to its ability for structural customization and printing scale. This process consists of depositing successive layers of concrete

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contouring an object without formwork, i.e., by direct concrete placement. Therefore, 3DCP is a free form buildup [1] and was proved on the rationalization of ecological materials [2, 3]. The basic physics of the full of manufacturing steps of 3DCP was detailed in the reference [4]. Despite its enormous progress, the process is still accompanied by several errors and failures. A specific constraint is that an angular toolpath needs to be implemented at the corner, which is typically rounded.

The trajectory planner sits between the G-code and the servomotor execution of the toolpath travel. The G-code file may contain a million lines in it because of how much path error is permissible, how long it is going to take, how smooth desired, the axes coupled and dynamic computation. Printing objects comes with the understanding of the extruder dynamics of the 3D printer and how planning the path [5]. Fleck et al. [6] observed an overflow at the corner because of trapezoidal velocity profiles employed through Marlin firmware [7]. This firmware involves a slam velocity at turn. Research work has been done on material extrusion-based additive manufacturing with investigating the print precision of cornering [8, 9]. Giberti et al. [10] proposed an algorithm for path planning to inspect the material extrusion rate strained by velocity pursuant of a toolpath over the execution of 3D printing process. Jin et al. [11] tried to smooth paths by decreasing the number of sharp corners using an implicit algorithm. Bos et al. [12] mentioned that square nozzles should be tangent to the toolpath when moving the printhead to overcome the twisting of filament. AlSakka et al. [13] identified the optimal deposition path of the 3DCP which reduce the weak bonds and preserve the buildability of the extruded material by using a discrete event model.

In complement to the physical printing, CFD modelling is able to reproduce the extrusion-deposition of the printable concrete and investigate some specific constraints to the physical print [14–17]. The angular toolpath planning in 3DCP is in its infancy. However, several CFD modelling studies have been performed in other 3D printing technologies, like fused deposition modeling. [18, 19] developed CFD models to investigate the underfill and overflow areas in the deposited corners, and investigated to smoothen the corners. Furthermore, Mollah et al. [20] investigated corner deposition for Bowden and Direct-drive extruders, where the Direct-drive extruder controls the amount of material extrusion near the corner. These models were developed using the Newtonian fluid model.

In the present study, a CFD model is developed to solve the Navier Stokes equations with the Bingham rheological model to simulate the mortar extrusion through a cylindrical nozzle. The simulations predict the evolution of the layer shape and the overfill area of a deposited mortar at a turn of 90° angle. Different printing ways are simulated to inquire the consequences of smoothing the extruder nozzle with a constant extrusion rate. The structure of the study is as follows. Section 16.2 describes the methodology of the study with the theoretical details, CFD model, and path planning strategy. The results are discussed in Sect. 16.3. Section 16.4 concludes the study.

16.2 Methodology

16.2.1 Governing Equations

The flow dynamics of fresh mortar is modelled by the Navier Stokes' equations for the non-Newtonian incompressible fluid with a constant density, formed by the following continuity and the momentum equations:

$$\nabla \cdot u = 0 \tag{16.1}$$

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u}\right) = -\nabla p + \rho \mathbf{g} + \nabla \cdot \boldsymbol{\sigma} \tag{16.2}$$

where $\boldsymbol{u}$ is the velocity vector field, $\rho = 2100$ kg/m^3 is the density, p is the pressure, $\boldsymbol{g}$ is the gravitational vector field acting downwards, σ is the constitutive stress tensor, and t is the time variable. The rheology of the fresh mortar can be modelled by the viscoplastic Bingham model [21]. This type of fluids flows when the shear stress exceeds the yield stress. The evolution of the constitutive stress tensor can be approximated with the Generalized Newtonian Fluid constitutive model:

$$\boldsymbol{\sigma} = 2\eta_{app}(\dot{\gamma})\mathbf{D} \tag{16.3}$$

where $\mathbf{D} = (\nabla \mathbf{u} + \nabla^T \mathbf{u})\big/2$ is the strain rate tensor, $\dot{\gamma} = \sqrt{2tr(\mathbf{D}^2)}$ is the shear rate, and $\eta_{app}(\dot{\gamma})$ is the apparent viscosity of the viscoplastic Bingham model. However, the apparent viscosity of the material increases to infinity when the applied stress is below the yield stress. To overcome this singularity issue of the Bingham fluid flow in the numerical simulation, the apparent viscosity is smoothed with the bi-viscosity method:

$$\eta_{app}(\dot{\gamma}) = \begin{cases} \eta_{app}^{MAX} & \text{for } \dot{\gamma} < \dot{\gamma}_c \\ \frac{\tau_Y}{\dot{\gamma}} + \mu_p & \text{for } \dot{\gamma} \geq \dot{\gamma}_c \end{cases} \tag{16.4}$$

where $\tau_Y = 630$ Pa is the yield stress, $\mu_p = 7.5$ Pa.s is the plastic viscosity, $\eta_{app}^{\text{MAX}} = \tau_Y\big/\dot{\gamma}_c + \mu_p$ is the maximum value of the smoothed apparent viscosity and $\dot{\gamma}_c = 0.01$ s^{-1} is the critical threshold of the shear rate [14]. For more details on the regularization methods of the Bingham constitutive equation, please see [22].

16.2.2 CFD Model

The physical problem was numerically simulated with the software FLOW-3D®, version 12.0 [23]. The Eqs. (16.1) and (16.2) are discretized by the finite volume

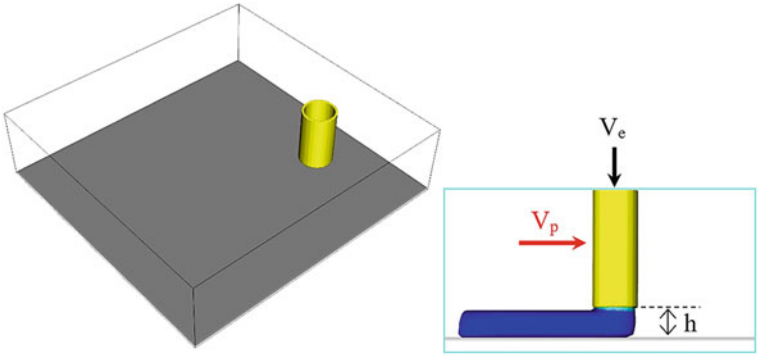

Fig. 16.1 Geometry of the CFD model (left) and a picture during the simulation (right)

method. The volume of fluid method [24] was used to capture the extrudate mortar sharp interfaces with the split Lagrangian method. The numerical domain was discretized with a uniform Cartesian non-conforming mesh. The cell size is 1 mm for each direction. A no slip boundary condition was applied to the nozzle wall and the build plate. The fresh mortar begins to flow through the extruder nozzle with a constant velocity and exit with a fully developed flow profile. The other domain boundaries are set as continuative boundary conditions. Intel® Xeon® W-2155 CPU 3.30 GHz 20 cores is used to execute all the numerical simulations. Each simulation takes around 2 days and 4.5 h. The geometry of the model consists of a cylindrical nozzle and a build plate. The nozzle comes with an inner diameter D = 25 mm and a thickness e = 2 mm, the numerical domain occupies a volume of 240 × 230 × 65 mm^3. The distance between the nozzle outlet and the build space is fixed at h = 12.5 mm. The printhead starts to move with a prescribed velocity when the mortar reaches the build plate (see Fig. 16.1).

16.2.3 Path Planning Strategy

The path planning of deposited corners consists of printing two segments with 90° turn without stopping the printhead. The servomotor of the 3D printer controls the displacement through a finite jerk and a linear acceleration limit. The jerk is the velocity leap that occurs instantly when the printhead begins an acceleration and a deceleration. Five depositing strategies of mortar layer with sharp and smooth corner have been simulated (see Fig. 16.2). The printhead moves with a maximum velocity Vp = 50 mm/s, decelerates and accelerates at the corner with a maximum value A_{max} = 30–50 mm/s^2. The deceleration time $\delta t = (t_{(s_1)out} - t_{(s_1)in}) = 1$ s and the extrusion velocity (V_e = 33.4 mm/s) are kept constant along the path except in the case (a) of the most used printing strategy in 3DCP, where the jerk is infinite because of the very small servo time (0.0001 s) at the acceleration transitions. A smoothing factor χ was used to smooth the printhead trajectory around the corner defined as the ration between the acceleration time limit $(t_{(s_1)out} - t_{(s_2)in})$ and the deceleration time δt required to decrease the velocity of the printhead until it stops in the first direction (X axis in our cases). S_1 and S_2 refer to the two segments respectively. The following Table 16.1 presents the path planning strategies that are numerically simulated.

16.3 Results and Discussion

The simulated results of the deposition flow are post-processed using FlowSight, version 12.0 and presented in Fig. 16.3. In the left side of the figure, we illustrate a perspective view of the different simulated strategies. The right side shows the ideal widths and the maximum flow depth of the different deposited layers. To exactly execute a toolpath with the G-code of the 3D printer, the extruder nozzle has to stop every time reaching a corner because in each segment of the toolpath, the extruder nozzle moves at a fixed ratio between the motors. When coming to a stop at the corner, the flow rate drew from the extruder nozzle and it will take a long time to deposit. Figure 16.3.a, b and c correspond to the sharp corner cases. The first strategy corresponds to an ideal model with an infinite jerk where we have to be aware because it could cause vibration and slam to the printhead and as a result a poor deposition could be obtained. It is observed that the selected jerk did not improve the print quality of the corner (see Fig. 16.3c). Furthermore, the mortar overflowed and crushed at the 90° turn. This is occurring because of the feed rate that has been kept constant and the increase of the extrusion pressure due to the slowing of the printhead while changing direction. In order to smooth corners, it is needed to act on the velocity, acceleration and jerk limits of the actuator of the 3D printer. Figure 16.3d shows an overflow at the exterior of the turn due to the corner shape and the acceleration time limit of 0.3 s. The accuracy print is obtained when the acceleration and the deceleration steps are blended instantly. Figure 16.3e shows a uniform corner bounded by an elliptical arc through the endpoints with a minimal overflow at the exterior of the turn despite of the smoothing of the toolpath. The observation is that when the printhead goes through the corner, the velocity is momentarily decelerated and accelerated. Therefore, a curvilinear path at the corner exterior is obtained.

The overflow and the underflow amount at the inside and the outside of the corner with 90° angle regarding to the ideal model (a) were quantified to analyze the deposition behavior when the printhead initiate a turn for the studied printing strategies (b), (c), (d), (e), and a printing strategy with a proportional extrusion velocity (see Fig. 16.4), where the deceleration/acceleration phases are blended (see Fig. 16.5).

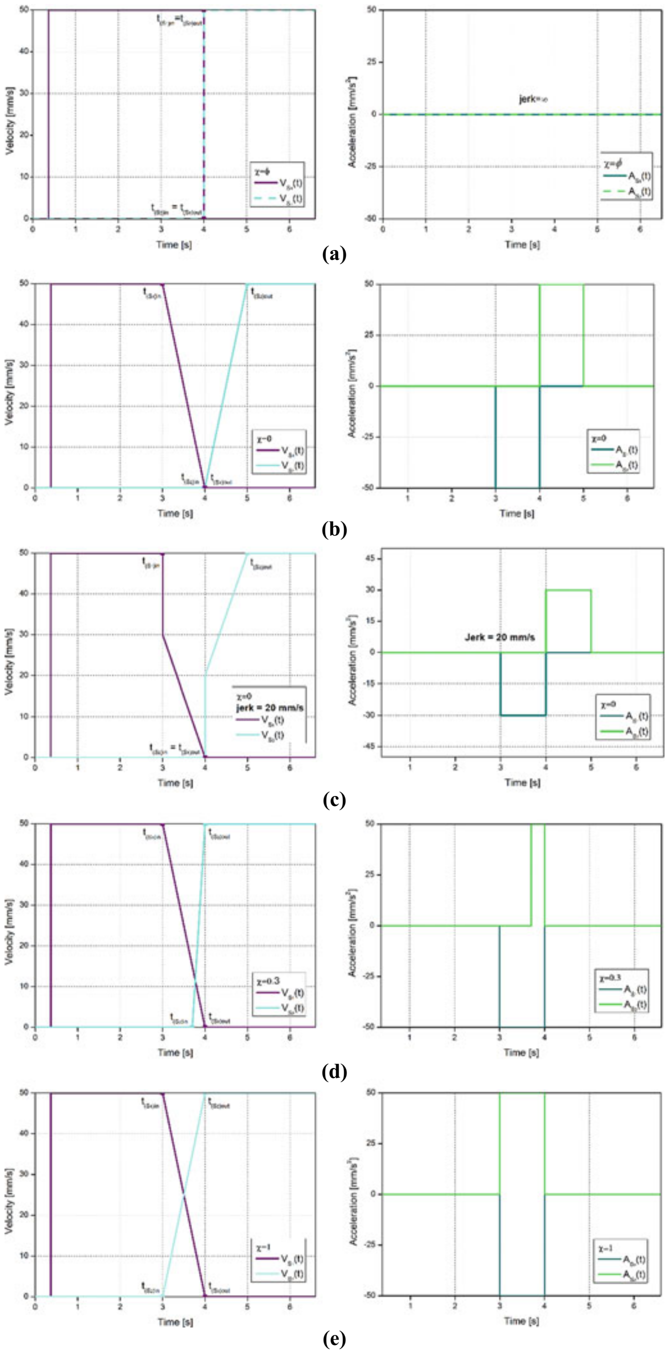

Fig. 16.2 Per-axis velocity profiles and there corresponding acceleration steps with different smoothing factor

Table 16.1 Path planning strategies cases numerically simulated

Corner type	Smoothing factor χ
Sharp corner	0 (infinite jerk)
	0 (without jerk)
	0 (jerk = 20 mm/s)
Smooth corner	0.3
	1

The blue bars correspond to a comparison between the case (c) and the printing strategy applying a jerk of 20 mm/s and a proportional extrusion velocity. The virtual printed layer for the sharp corner strategy applying a jerk of 20 mm/s was noticeably improved because of the synchronization of the extrusion rate at the direction change of the printhead (blue bar at the top). As illustrated on the graphs, the deposited material largely overflowed at the exterior side of the turn compared to the interior side for all the toolpath planning scenarios. This is because the double deposition occurs at the inner side of the turn, one from the pre-corner strand and the other from the post-corner layer. In addition, the toolpath of the printhead was fitted at the turn to have a curvature line without reaching the sharp corner. Therefore, the precision of the print quality depends on the way we program the print settings.

16.4 Conclusions

A CFD model was developed to investigate different toolpath strategies to assess the corner precision in 3DCP. The rheology of the printable mortar is approximated with the Generalized Newtonian constitutive model using the Bingham rheological fluid model. A 90° turn was investigated. It was found that the printhead should be planned to not slam or slow to a stop for good print quality of corners. It was observed that the tested case involving a jerk of 20 mm/s and an extrusion rate synchronized with the nozzle speed reduces the swelling of the strand at the corner. In fact, a commensurate extrusion rate with the printing velocity should generate a uniform corner in the case of the 90° turn; however, it could create a major swelling of the strand in the case of smaller angles. The CFD model provides a powerful tool that could be utilized to investigate different angular turns in 3DCP. This will minimize the number of prototypes and experimental tests required to implement toolpath strategies for corners with high precision.

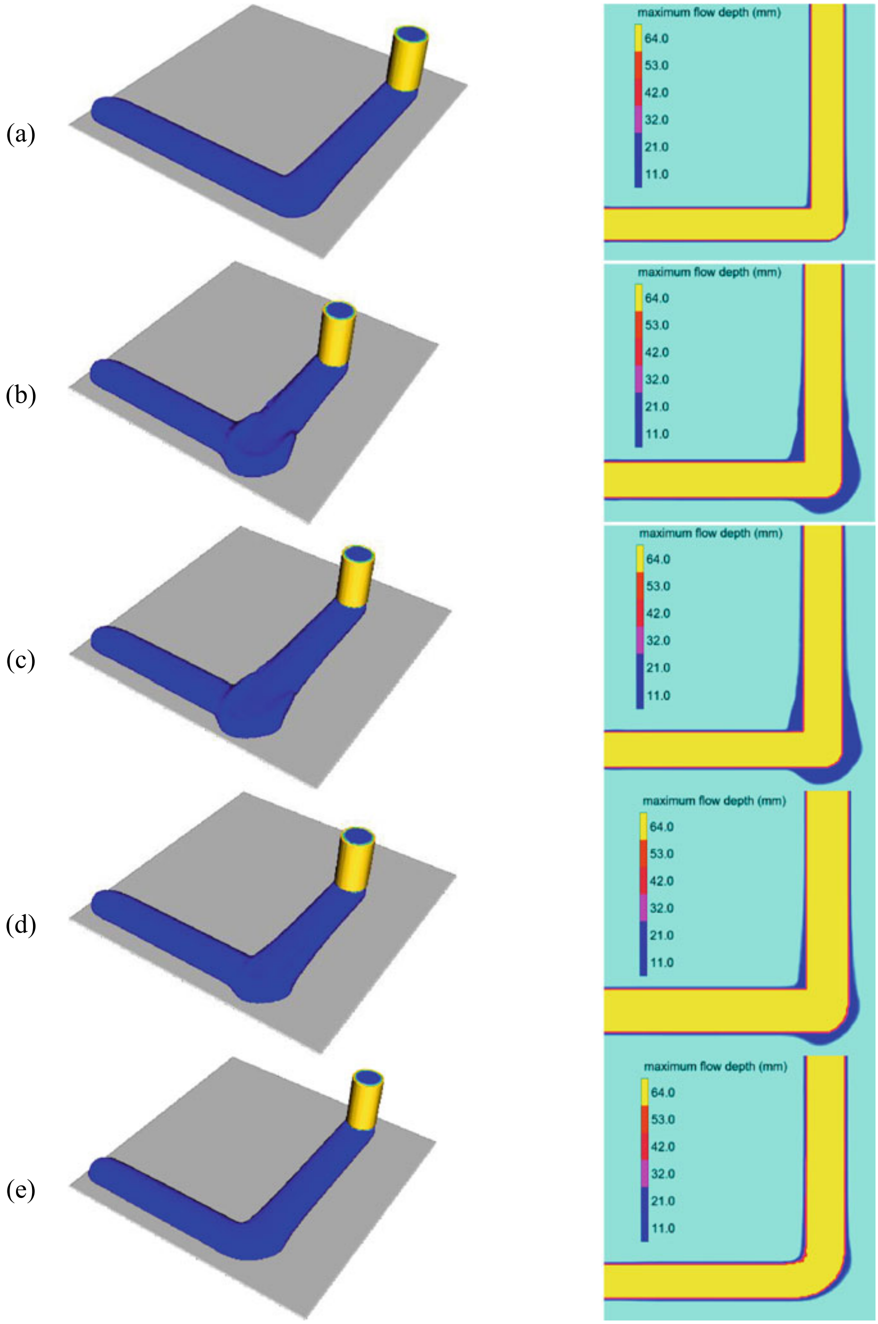

Fig. 16.3 Simulations of mortar extrusion along a toolpath with direction change of 90° using five deposition strategies

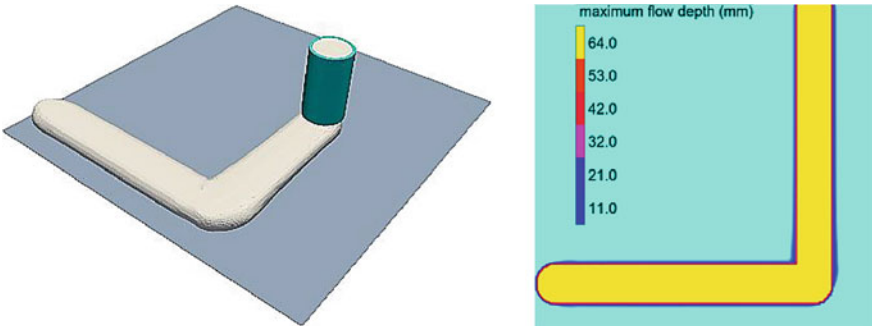

Fig. 16.4 Printing strategy applying a jerk of 20 mm/s and a synchronized extrusion velocity with the printing veocity

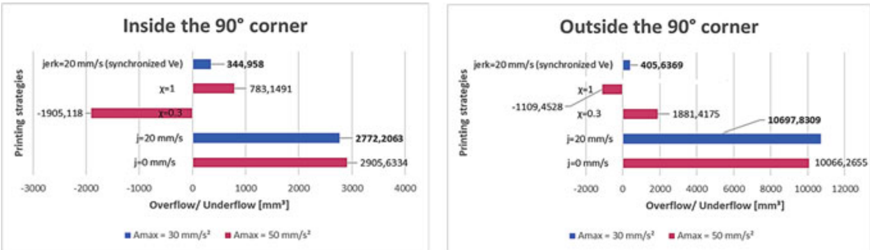

Fig. 16.5 Underflow/Overflow amount of the 90° corner print, **a** Inside the 90° corner **b** Outside the 90° corner

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